

TECHNICAL DATA / EXTRA INFORMATION

Unibar System

Generic technical datasheet for mechanical theatre equipment compliance records

Important scope note

This document is a generic technical datasheet template for a Unibar or similar motorised bar / internally wired bar lifting system. It is not a substitute for the manufacturer manual, original design information, commissioning records, electrical certification, site risk assessment, or a competent person's thorough examination report.

1. Equipment overview

A Unibar system is a theatre or studio lifting arrangement used to raise, lower, support or position a bar, internally wired bar, lighting bar, track or similar production item. The exact design may vary by manufacturer and installation. Some systems are motorised, some include internally wired outlets, and some form part of a wider stage machinery or venue flying system.

The system should be assessed as a complete arrangement. This includes the bar, lifting machinery, wire ropes or belts, pulleys, drum or winding components, supports, brackets, fixings, control system, limits, emergency stop devices, load path and supporting structure.

Summary	
Typical use	Raising, lowering or supporting lighting bars, internally wired bars, track, scenic items or other approved production equipment.
Primary risk areas	Overhead suspended load, moving load, electrical services, control failure, rope/belt deterioration, poor support, overload and uncontrolled movement.
Portal equipment class	Powered lifting machine / structural support depending on the final installed arrangement.
Not covered by this template	Manufacturer-specific capacities, wiring diagrams, control schematics, software settings or commissioning values unless added by a competent person.

2. Identification and site details

Asset record	
Venue / site	To be completed
Area / room / stage	To be completed
Equipment ID / asset number	To be completed
Manufacturer	To be completed
Model / type	To be completed
Serial number	To be completed
Year of manufacture / installation	To be completed
Installer / commissioning body	To be completed
Control location / panel ID	To be completed
Related drawings / manuals	To be completed

3. Technical data to confirm

Technical information	
Safe Working Load / WLL	To be confirmed from manufacturer information, system design and current inspection records.
Bar length / type	To be completed - include whether plain, internally wired, aluminium, steel, track, or other.
Travel distance	To be completed
Lifting medium	Wire rope / chain / belt / other - to be completed
Drive arrangement	Motor, gearbox, drum, hoist, winch, drive shaft or other - to be completed
Speed / duty cycle	To be completed from manufacturer or system documentation.
Control system	Pendant, wall panel, desk, group control, dead-man control, key switch or other - to be completed.
Safety devices	Upper/lower limits, emergency stop, brake, overload

	detection, slack rope/belt detection, load monitoring, key switch or other - to be completed.
Power supply	To be completed by competent electrical person.
Inspection interval	To follow LOLER/PUWER requirements, written scheme, manufacturer instructions and site risk assessment.
Known limitations	To be completed - include any loading, travel, operational, environmental or access restrictions.

Loading note

Do not assume that the bar capacity is the same as the lifting machinery capacity. The lowest rated part of the complete load path controls the usable capacity. Point loads, distributed loads, support spacing and electrical outlet loading should all be considered.

4. Applicable compliance context

LOLER: A Unibar system used to lift, lower or suspend loads is likely to fall within LOLER. The system should be strong and stable, suitable for the task, marked where appropriate, and subject to competent examination and inspection as required.

PUWER: The system is work equipment and should be suitable, maintained, inspected where deterioration could create risk, and used only by people with adequate information, instruction and training.

BS EN 17206: Where the system forms part of entertainment machinery or stage machinery, BS EN 17206 should be considered for safety functions, control systems, prevention of uncontrolled movement, commissioning, verification, documentation and maintenance.

ABTT Technical Standards: ABTT guidance supports recognised theatre-industry good practice for installation, inspection, maintenance, documentation and competent operation in places of entertainment.

5. Installation and support considerations

- Confirm that the supporting structure is suitable for the imposed loads, including dynamic effects and any load concentration.
- Confirm that suspension points, brackets, fixings, beams, drop rods or supports match the original design or have been assessed by a competent person.
- Check that the bar or load carrier remains straight, correctly aligned and free from excessive deflection throughout travel.
- Confirm that the lifting medium is correctly routed, seated, tensioned and compatible with the drum, pulleys, guides or sheaves.
- Where the system includes electrical outlets or internally wired sections, confirm that electrical containment and strain relief are protected from mechanical damage.
- Confirm that access for inspection, maintenance and emergency recovery is available and safe.

6. Operating considerations

- Only competent and authorised persons should operate the system.
- Before movement, check that the travel path is clear and that the load is within the approved operating limits.
- Do not use the system for lifting people unless specifically designed, documented and approved for that purpose.
- Do not bypass emergency stops, limit switches, interlocks, brakes or load monitoring devices.
- Do not alter loads, add concentrated loads or change the bar configuration without assessment.
- Stop operation immediately if there is abnormal noise, uneven movement, drift, snatching, slipping, poor winding, visible damage or unexpected control behaviour.

7. Inspection checklist

Typical checks

No.	Check item	Status / notes
1	Confirm identification, SWL/WLL and operating limits are known and visible or otherwise documented.	☐ OK ☐ Defect ☐ N/A Notes:
2	Inspect the bar, track or load carrier for bending, twisting, dents, cracks, corrosion, deformation or damage.	☐ OK ☐ Defect ☐ N/A Notes:
3	Inspect lifting machinery, motor, gearbox, drum, winch, hoist, bearings and housings for damage, looseness, leaks or abnormal wear.	☐ OK ☐ Defect ☐ N/A Notes:
4	Inspect wire ropes, chains, belts or lifting media for wear, damage, corrosion, broken wires, kinks, flattening, contamination or incorrect seating.	☐ OK ☐ Defect ☐ N/A Notes:
5	Check pulleys, sheaves, guides and diverters for alignment, free rotation, secure fixing and wear.	☐ OK ☐ Defect ☐ N/A Notes:
6	Check terminations, sockets, clamps, thimbles, anchor points and end fixings for correct fitting and damage.	☐ OK ☐ Defect ☐ N/A Notes:

7	Check suspension points, beams, brackets, fixings, drop rods and supporting structure for movement, corrosion, cracking or distortion.	☐ OK ☐ Defect ☐ N/A Notes:
8	Test controls, emergency stop devices, limits, brakes and safety functions where appropriate and safe to do so.	☐ OK ☐ Defect ☐ N/A Notes:
9	Check cable management, flexible cables, sockets, glands and electrical containment for damage, strain or snagging.	☐ OK ☐ Defect ☐ N/A Notes:
10	Operate through the safe working travel and check for smooth movement, correct stopping, no drift, no jamming and no abnormal noise.	☐ OK ☐ Defect ☐ N/A Notes:
11	Confirm documentation, inspection reports, maintenance records and defect records are available and current.	☐ OK ☐ Defect ☐ N/A Notes:
12	Confirm that any modification, repair or change of use has been assessed and recorded before the system is returned to service.	☐ OK ☐ Defect ☐ N/A Notes:

8. Defects requiring action

- Missing or unclear SWL/WLL, asset identification or operating limitations.
- Any cracked, bent, distorted, corroded or damaged structural component.
- Damaged, worn, kinked, crushed, contaminated or incorrectly routed wire rope, belt, chain or lifting medium.
- Poor winding on a drum, rope riding, crossing, bunching, slipping or rubbing.
- Defective brakes, limits, emergency stops, controls, interlocks, overload devices or safety functions.
- Uncontrolled movement, drift, uneven travel, abnormal noise, vibration or snatching.
- Loose fixings, damaged brackets, movement in supports or signs of structural distress.
- Damaged electrical outlets, exposed conductors, strained cables, missing covers or damaged containment.
- Unauthorised modification, non-standard repairs or undocumented changes to the system.

Defect response

If a defect could affect safe use, take the system out of service, prevent unauthorised operation and refer it to

a competent person. Do not return the system to service until the defect has been assessed, corrected and recorded.

9. Records and document links

Associated records	
Manufacturer manual	To be attached or referenced
Original design / calculations	To be attached or referenced
Electrical certificate / test records	To be attached or referenced
Commissioning certificate	To be attached or referenced
LOLER thorough examination report	To be attached or referenced
PUWER inspection / maintenance record	To be attached or referenced
Defect / repair history	To be attached or referenced
Site-specific safe system of work	To be attached or referenced

10. Portal summary wording

Suggested portal note

Use this datasheet as a generic technical record for a Unibar system or similar motorised theatre lifting bar. Add manufacturer-specific data, site details, SWL/WLL, control information, inspection records and any limitations before relying on it for inspection or maintenance planning.

Document status: generic editable template - complete site-specific details before use



RAISE & LOWER SUSPENSION BARS QUICKLY & EASILY





Unibar®

A cost-effective alternative to electrically operated pile-wind winches and manual winch and hemp sets.

The Unibar® system allows for the quick and easy controlled raising and lowering of suspension bars for general lifting purposes: stage scenery, curtains, speakers and lighting equipment. Unibar® is a flexible user-friendly unit, designed to be installed on to any style of fixing or structure such as 'Universal Beams', Trussing and Channels.

Unibar® is a cost effective alternative to electrically operated pile-wind winches while also providing an alternative to manual winch and hemp sets. Unibar® is ideal for schools, colleges, sports halls, theatres, amateur dramatic and touring applications.

- Suitable for lifting of stage scenery, curtains, speakers and lighting equipment
- User-friendly unit, designed to be installed onto any style of fixing or structure
- A cost-effective alternative to electrically operated pilewind winches

See the Unibar® installation guide for loads and recommendations. Please note: a competent person must ensure installation integrity with existing structures.

Features and benefits

The Unibar® has been designed to support up to five suspension cables with standard units available in 3 line, 4 line and 5 line configurations.

Adjustable working top and bottom electrical limits contained within the Unibar® motor control determine the top and bottom limits of travel of the suspension bar.

The Unibar® is designed to allow two units to be positioned back to back a 320mm centres.

Up and down emergency stop control of the Unibar® is via a hanging hand pendant control, or a wall mounted panel.

The Chain Hoist has an 8:1 safety factor. If a 10:1 safety factor is required, the maximum safe working load would be 320kg.

For added safety the start stop control is key operated.

The Unibar® is designed with all of the suspension fixings level and uniform fixing centres, allowing quick and easy fitting in any situation.

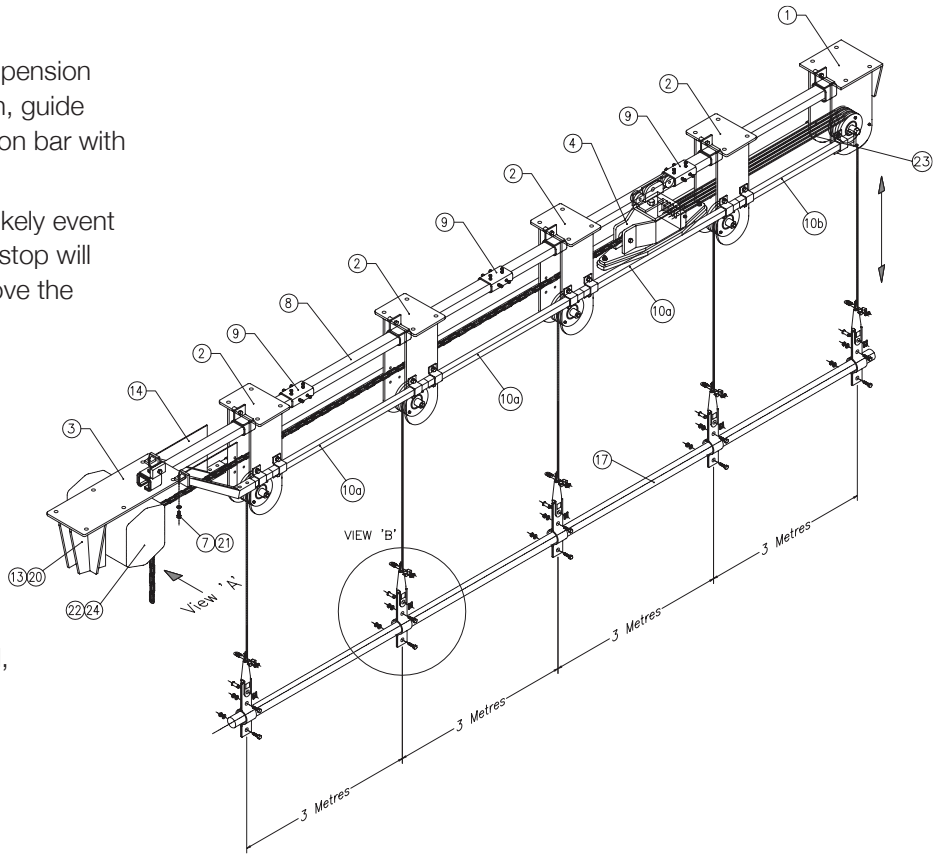
The Unibar® hauling chain runs out of the chain hoist and is attached to a guided horizontally travelling clew.

Attached to the clew is also a series of suspension cables which pass around ball raced return, guide and drop pulleys terminating at a suspension bar with height adjustable clamps.

As an additional safety measure, in the unlikely event of upper limit failure an independent chain stop will prevent the suspension bar from rising above the intended position.

Each chain hoist has as standard:

- Adjustable internal limits to control the extremities of travel.
- Wear resistant overload slip clutch; with asbestos free lining.
- Heat treated gears in oil bath.
- Wear resistant brake (no resetting necessary).
- Load chain to Din 5684 cadmium plated, grade 8.
- Motor protection to IP54.
- Motor insulation to grade F.



Unibar® range

Product no.	Lines/ Unit	Bar Length (m)	Chain Hoist Model	Capacity (kg)	Voltage	Duty Cycle	Motor Power (kw)	Full Load	Lifting Speed (m/min.)	Height of Lift (m)	Self Weight (kg)
3L3250	3	8	GCH500/IN	250	415/3ph	60%	0.82	2.1A	9	5.9	119
4L3250	4	10	GCH500/IN	250	415/3ph	60%	0.82	2.1A	9	8.9	147
5L3250	5	12	GCH500/IN	250	415/3ph	60%	0.82	2.1A	9	11.9	177
3L1250	3	8	GCH500/IN	250	220/1	40%	0.42	4.9A	9	5.9	119
4L1250	4	10	GCH500/IN	250	220/1	40%	0.42	4.9A	9	8.9	147
5L1250	5	12	GCH500/IN	250	220/1	40%	0.42	4.9A	9	11.9	177
3L3500	3	8	GCH500/IN	400	415/3ph	30%	0.82	2.1A	9	5.9	119
4L3500	4	10	GCH500/IN	400	415/3ph	30%	0.82	2.1A	9	8.9	147
5L3500	5	12	GCH500/IN	400	415/3ph	30%	0.82	2.1A	9	11.9	177
3L1500	3	8	GCH500/IN	400	220/1	25%	0.42	6.4A	4.5	5.9	119
4L1500	4	10	GCH500/IN	400	220/1	25%	0.42	6.4A	4.5	8.9	147
5L1500	5	12	GCH500/IN	400	220/1	25%	0.42	6.4A	4.5	11.9	177



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